.1 For constant -speed marine engines for ship main propulsion, including diesel-electric drive, test cycle E2 shall be applied.

.2 For controllable-pitch propeller sets test cycle E2 shall be applied.

.3 For propeller-law-operated m ain and propeller-law-operated auxiliary engines the test cycle E3 shall be applied.

.4 For constant-speed auxiliary engines test cycle D2 shall be applied.

.5 For variable-speed, variable-load auxiliary engines, not included above, test cycle C1 shall be applied.

Test cycle for constant-speed main propulsion application
(including diesel-electric drive and all controllable-pitch propeller installations)

Test cycle type
E2
Speed 100% 100% 100% 100%
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15

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(including diesel-electric drive and all controllable-pitch propeller installations)

Test cycle type
E2
Speed 100% 100% 100% 100%
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15

Test cycle for propeller-law-operated main and propeller-law-operated auxiliary engine application

Test cycle type
E3
Speed 100% 91% 80% 63%
Power 100% 75% 50% 25%
Weighting factor 0.2 0.5 0.15 0.15

Test cycle for constant-speed auxiliary engine application

Test cycle type
D2
Speed 100% 100% 100% 100% 100%
Power 100% 75% 50% 25% 10%
Weighting factor 0.05 0.25 0.3 0.3 0.1

Test cycle for variable-speed and variable-load auxiliary engine application

Test cycle type
C1
Speed Rated Intermediate Idle
Torque 100% 75% 50% 10% 100% 75% 50% 0%
Weighting factor 0.15 0.15 0.15 0.1 0.1 0.1 0.1 0.15

In the case of an engine to be certified in accordance with paragraph 5.1.1 of regulation 13, the specific emission at each individual mode point shall not exceed the applicable NO x emission limit value by more than 50% except as follows:

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the specific emission at each individual mode point shall not exceed the applicable NO x emission limit value by more than 50% except as follows:

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.1 The 10% mode point in the D2 test cycle.
.2 The 10% mode point in the C1 test cycle.
.3 The idle mode point in the C1 test cycle.

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Appendix III

Criteria and procedures for the designation of emission control areas
(regulations 13.6 and 14.3)

1 Objectives

1.1 The purpose of this appendix is to provide Parties with the criteria and procedures for formulating and submitting proposals for the designation of emission control areas and to set forth the factors to be considered in the assessment of such proposals by the Organization.

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formulating and submitting proposals for the designation of emission control areas and to set forth the factors to be considered in the assessment of such proposals by the Organization.

1.2 Emissions of NOx, SOx and particulate matter from ocean-going ships contribute to ambient concentrations of air pollution in cities and coastal areas around the world. Adverse public health and environmental effects associated with air p ollution include premature mortality, cardiopulmonary disease, lung cancer, chronic respiratory ailments, acidification and eutrophication.

1.3 An emission control area should be considered for adoption by the Organization if supported by a demonstrated need to prevent, reduce and control emissions of NOx or SOx and particulate matter or all three types of emissions (hereinafter emissions) from ships.

2 Process for the designation of emission control areas

2.1 A proposal to the Organization for the designation of an emission control area for NOx or SOx and particulate matter or all three types of emissions may be submitted only by Parties. Where two or more Parties have a common interest in a particular area, they should formulate

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or SOx and particulate matter or all three types of emissions may be submitted only by Parties. Where two or more Parties have a common interest in a particular area, they should formulate a coordinated proposal.

2.2 A proposal to designate a given area as an emission control area should be submitted to the Organization in accordance with the rules and procedures established by the Organization.

3 Criteria for designation of an emission control area

3.1 The proposal shall include:

.1 a clear delineation of the proposed area of application, along with a reference chart on which the area is marked;

.2 the type or types of emission(s) that is or are being proposed for control (i.e. NOx or SOx and particulate matter or all three types of emissions);

.3 a description of the human populations and environmental areas at risk from the impacts of ship emissions;

.4 an assessment that emissions from ships operating in the proposed area of application are contributing to ambient concentrations of air pollution or to adverse environmental impacts. Such assessment shall include a description of the impacts of the relevant emissions on human health and the

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application are contributing to ambient concentrations of air pollution or to adverse environmental impacts. Such assessment shall include a description of the impacts of the relevant emissions on human health and the environment, such as adverse impacts on terrestrial and aquatic ecosystems, areas of natural productivity, critical habitats, water quality, human health, and areas of cultural and scientific significance, if applicable.

The sources of relevant data including methodologies used shall be identified;

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.5 relevant information, pertaining to the meteorological conditions in the proposed area of application, to the human populations and environmental areas at risk, in particular prevailing wind patterns, or to topographical, geological, oceanographic, morphological or other conditions that contribute to ambient concentrations of air pollution or adverse environmental impacts;

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areas at risk, in particular prevailing wind patterns, or to topographical, geological, oceanographic, morphological or other conditions that contribute to ambient concentrations of air pollution or adverse environmental impacts;

.6 the nature of the ship traffic in the proposed emission control area, including the patterns and density of such traffic;